

Algebra II with Trigonometry

Chapter 5.1

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

1. **Degrees to Radians** Convert 150° to radians.
2. **Radians to Degrees** Convert $\frac{3\pi}{4}$ radians to degrees.
3. **Positive or Negative** If an angle rotates clockwise from its initial side, is its measure positive or negative?
4. **Coterminal Angle Check** Are 45° and 405° coterminal?
5. **Another Coterminal Check** Are $\frac{\pi}{3}$ and $\frac{7\pi}{3}$ coterminal?
6. **Quick Arc Length** In a circle of radius 5, what is the arc length subtended by an angle of 2 radians?
7. **Find the Angle from Arc Length** In a circle of radius 4, an arc has length 12. What is the central angle in radians?
8. **Quick Sector Area** What is the area of a sector with radius 6 and central angle 2 radians?
9. **Angular Speed** A point sweeps out 8π radians in 4 seconds. What is its angular speed?
10. **Linear Speed from Angular Speed** A point moves on a circle of radius 3 with angular speed 2 radians per second. What is its linear speed?